CSE4078 SPRING 2026 TERM PROJECT DESCRIPTION

For the Spring 2026 term project, you will work in groups to evaluate and fine-tune small open-source Large Language Models (LLMs) for Turkish. In order to ensure fair comparison across groups, all groups will use the same public test corpus and the same primary supervised fine-tuning (SFT) corpus. Project has only two graded deadlines: a progress presentation and a final submission with a final presentation.

Project Objective

The main goal of the project is to measure the Turkish performance of small LLMs, apply a focused fine-tuning process, and analyze whether fine-tuning improves performance on a shared public Turkish test set.

Each group is expected to:

1. Select two small open-source LLMs in the approximately 0.3B to 4B parameter range.
2. Evaluate both base models on the assigned common public Turkish test corpus.
3. Compare the baseline results and select one model for fine-tuning.
4. Fine-tune the selected model using the common SFT corpus.
5. Re-evaluate the fine-tuned model on the same public Turkish test corpus.
6. Analyze strengths, weaknesses, and typical error patterns before and after fine-tuning.

To keep the project manageable, each group is required to fine-tune only one model. Parameter-efficient fine-tuning methods such as LoRA or QLoRA are strongly recommended.

Mandatory Public Datasets

All groups must use the following datasets.

1. Mandatory SFT Corpus

All even numbered groups use: Renicames/turkish-law-chatbot

All odd numbered groups use: alibayram/doktorsitesi

- Renicames/turkish-law-chatbot, which has 14.9k rows with a 13.4k train split and a 1.5k test split, under Apache-2.0.
- alibayram/doktorsitesi, which contains 167,732 Turkish medical question-answer pairs with fields such as question_content and question_answer. Focus on questions including following medical fields: beyin-ve-sinir-cerrahisi, uroloji, ortopedi-ve-travmatoloji, dahiliye-ve-ic-hastaliklari*, genel-cerrahi, kulak-burun-bogaz-hastaliklari, fiziksel-tip-ve-rehabilitasyon, kardiyoloji, kalp-damar-cerrahisi. If you like you can sample but your training split can not be less then 15 thousand instances and your test set can not be less then 1500 instances. Class (medical fields) distribution must match the training set.

SFT Data Usage Rule

Use training set split for fine-tuning. The test split corpus must remain completely unseen during training. The goal is to evaluate whether fine-tuning improves general Turkish benchmark performance, not to memorize the test set. The SFT dataset is large enough to support this requirement.

GitHub Repository Requirement

Each group must create a GitHub repository named:

CSE4078S26_GrpX

Replace X with your group number.

All group members must actively contribute to the repository. Commit history may be considered as part of the evaluation of individual contribution. Your repository should include your evaluation scripts, fine-tuning scripts, inference scripts, analysis code, and a clear README explaining how to run the code.

Deadline 1: Progress Presentation
Due: 2026-05-18 Monday 09:00am
Presentation Week: 14

By this deadline, each group is expected to complete the baseline evaluation stage.

Before Deadline 1, each group should:

1. Select two small open-source LLMs.
2. Download the mandatory test corpus and evaluate both models on it.
3. Download the mandatory SFT corpus and prepare the SFT subset you plan to use.
4. Decide which one of the two models you will fine-tune.
5. Prepare a short progress presentation.

Files required for Deadline 1

1. Progress presentation in PDF format
   Filename:
   CSE4078S26_GrpX_ProgressPresentation.pdf

The presentation should explain:

- the two models you selected,
- your baseline evaluation setup,

- benchmark results for both models,
- your selected fine-tuning model,
- your plan for using the SFT corpus.

2. Progress summary in PDF format
   Filename:
   CSE4078S26_GrpX_ProgressSummary.pdf

This short document should include:

- group number and member names,
- selected baseline models,
- test corpus used,
- SFT corpus used,
- summary table of baseline results,
- model selected for fine-tuning.

Deadline 2: Final Submission and Final Presentation
Due: 2026-06-08 Monday 09:00am
Presentation Week: 17

By this deadline, each group must complete fine-tuning, final evaluation, error analysis, and academic reporting.

Files required for Deadline 2

1. Final report in IEEE proceedings format using LaTeX
   Filename:
   CSE4078S26_GrpX_FinalReport.pdf

The final report must be written in a format similar to an IEEE conference proceedings paper, prepared in LaTeX, and must be at most 10 pages long.

The report must include the following sections:

- Abstract
- Introduction
- Related Work
  (may be included inside the Introduction if you prefer)
- Experiment Setup
- Experiment Results
- Conclusion

The report should clearly describe:

- the selected base models,

- the SFT corpus
- preprocessing and deduplication steps,
- fine-tuning method and hyperparameters,
- evaluation method on the common test corpus,
- before/after results,
- error analysis and conclusions.

2. Final presentation in PDF format
   Filename:
   CSE4078S26_GrpX_FinalPresentation.pdf

The final presentation should summarize:

- the selected model,
- the SFT corpus and preparation,
- fine-tuning setup,
- benchmark results before and after fine-tuning,
- example outputs and error analysis,
- main conclusions.

3. Training and inference code downloaded from GitHub as a ZIP file
   Filename:
   CSE4078S26_GrpX_Code.zip

This ZIP file should contain:

- training code,
- inference code,
- evaluation code,
- preprocessing scripts,
- README file,
- any helper scripts needed to reproduce the results.

4. Dataset links document
   Filename:
   CSE4078S26_GrpX_DatasetLinks.pdf

This document must list all datasets used in the project and include:

- dataset name,
- dataset identifier,
- source,
- short description,
- how it was used in the project.

Overall Grading Breakdown

Deadline 1

- Progress Presentation: 25%
- Progress Summary and Baseline Evaluation Quality: 10%

Deadline 2

- Final Report: 15%
- Final Presentation: 40%
- Code Quality, Reproducibility, and Submission Completeness: 10%

Total: 100%

Important Rules

1. Each group must evaluate two base models first, then fine-tune only one selected model.
2. The benchmark test set must not be used in training or adaptation.
3. Model size should remain approximately in the 0.3B to 4B range.
4. Your work must be reproducible.
5. All group members are expected to contribute actively.
6. The final report must follow IEEE-style proceedings formatting and be written in LaTeX.
7. Only one student from each group should upload the required files to Google Classroom unless stated otherwise.